

IMEN266 Operations Research II

Concept & Proof Companion — Chapter 7 (Renewal theory)

[GAP-FILL version — fill the blanks yourself first]

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How to use this document

This companion *replaces your class notes*: every definition and theorem from the Ch. 7 slides (and the derivations developed in class) is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these eight proofs and their checkpoints are exactly the kind of reasoning the exams reward. Items marked **(final-style)** mirror questions from past exams: the renewal function of a hyperexponential distribution (Proof 2), the fraction of time the age process is below a constant via the alternating renewal process (Proof 8), and the five class problems themselves (Proofs 3, 5, 6, 8), which are natural exam material.

1 Concept digest

1.1 Renewal processes (slides pp. 1–6/54)

Let $X_1, X_2, \dots$ be iid nonnegative random variables with distribution F , $F(0) = P(X_n = 0) < 1$, and mean $\mu = E[X_1] \in (0, \infty]$. Put $S_0 = 0$, $S_n = X_1 + \dots + X_n$ (the time of the n -th **renewal**) and

$$N(t) = \max\{n : S_n \leq t\} \quad (\text{renewals in } (0, t]).$$

$\{N(t), t \geq 0\}$ is a **renewal process**. The Poisson process is the special case $F = \exp(\lambda)$ — the only one that “restarts” at every time; a general renewal process restarts only at its renewal epochs (slide p. 4/54). Key identity and consequences (Proof 1):

$$N(t) \geq n \iff S_n \leq t, \quad P(N(t) = n) = F_n(t) - F_{n+1}(t), \quad F_n = \text{cdf of } S_n,$$

$N(t) < \infty$ a.s. and $m(t) = E[N(t)] < \infty$ for every t . For exponential gaps $S_n \sim \text{Erlang}(n, \lambda)$ and the formula returns $e^{-\lambda t}(\lambda t)^n/n!$. (*verified in notebook: CP9 warm-up, §A*)

1.2 The renewal function and the renewal equation (slides pp. 7–12/54)

The **renewal function** $m(t) = E[N(t)] = \sum_{n \geq 1} F_n(t)$ (tail-sum formula) solves the **renewal equation**

$$m(t) = F(t) + \int_0^t m(t-x) dF(x) \quad (\text{condition on } X_1; \text{Proof 2}).$$

Uniform(0, 1) gaps: $m'(t) = 1 + m(t)$, $m(0) = 0$, so $m(t) = e^t - 1$ on $[0, 1]$. In general m is computed on a grid, by Monte-Carlo, or by Laplace transforms: with $\tilde{F}(s) = \int_0^\infty e^{-st} dF(t)$, $\tilde{m}(s) = \tilde{F}(s)/(1 - \tilde{F}(s))$ (Proof 2, final-style: hyperexponential F). The renewal function determines F and satisfies $m(t) = t/\mu + (\sigma^2 - \mu^2)/(2\mu^2) + o(1)$ when F is non-lattice with $\sigma^2 = \text{Var}[X] < \infty$ (not proved here). (*verified in notebook: CP9 §B: grid solver, Monte-Carlo and closed forms agree*)

1.3 Limit theorems (slides pp. 13–22/54)

Theorem (rate). $N(t)/t \rightarrow 1/\mu$ almost surely as $t \rightarrow \infty$ (Proof 3; $1/\mu = 0$ if $\mu = \infty$).

Elementary renewal theorem. $m(t)/t \rightarrow 1/\mu$ (Proof 3; needs Wald's equation).

Both say the same thing in two senses: the long-run rate of renewals is $1/\mu$. Only the *mean* of F sets the rate; the shape of F sets how fast $N(t)/t$ settles down (the fluctuation of $N(t)$ has standard deviation $\approx \sigma\sqrt{t/\mu^3}$). Applications: a battery with uniform(50, 100) lifetimes is replaced at rate 1/75 per hour (CP9 #1); in a single-server bank with Poisson(λ) arrivals that are accepted only when the server is free (service mean μ_G), cycles are “service + idle $\exp(\lambda)$ ” and the entering rate is $\lambda/(1 + \lambda\mu_G)$, the proportion entering $1/(1 + \lambda\mu_G)$ (CP9 #2; only the mean service time enters, the Poisson arrivals are essential). (*verified in notebook: CP9 §C–§D, #1, #2*)

1.4 Stopping times and Wald's equation (slides pp. 23–29/54)

A nonnegative integer-valued N is a **stopping time** for the sequence $X_1, X_2, \dots$ if the event $\{N = n\}$ is determined by $X_1, \dots, X_n$ (it may use the past, never the future). **Wald's equation:** if the X_n are iid with $E|X| < \infty$ and N is a stopping time with $E[N] < \infty$, then

$$E\left[\sum_{n=1}^N X_n\right] = E[N] E[X] \quad (\text{Proof 4}).$$

Both hypotheses matter: for the symmetric random walk and $N = \min\{n : S_n = 1\}$, N is a stopping time but $E[N] = \infty$ and $E[S_N] = 1 \neq E[N]E[X]$; for cards dealt without replacement the terms are not iid. For a renewal process $N(t)$ is *not* a stopping time ($\{N(t) = n\} = \{S_n \leq t < S_{n+1}\}$ looks at X_{n+1}), but $N(t) + 1$ is, hence the corollary $E[S_{N(t)+1}] = \mu(m(t) + 1)$. (*verified in notebook: CP9 §E*)

1.5 Renewal reward processes (slides pp. 30–34/54)

Let the pairs (X_n, R_n) be iid — the reward R_n of the n -th cycle may depend on *its own* length — with $E|R| < \infty$, $E[X] < \infty$, and let $R(t) = \sum_{n=1}^{N(t)} R_n$ be the reward earned by time t . Then

$$\frac{R(t)}{t} \rightarrow \frac{E[R]}{E[X]} \quad \text{a.s.} \quad \text{and} \quad \frac{E[R(t)]}{t} \rightarrow \frac{E[R]}{E[X]} \quad (\text{Proof 5}) :$$

the long-run reward rate is a ratio of means, never $E[R/X]$. Recipe: (1) find a renewal epoch (the system restarts probabilistically); (2) $E[X]$; (3) $E[R]$; (4) divide. CP10 #3 (door with a lockout t , Poisson(λ) arrivals, $\exp(\mu)$ service, costs c at the door and K at the busy server): cycles $X = t + \exp(\lambda)$, cost rate $(\lambda tc + Ke^{-\mu t} \lambda / (\lambda + \mu)) / (t + 1/\lambda)$. (*verified in notebook: CP10 warm-up, #3*)

1.6 Age, remaining time, and the inspection paradox (slides pp. 35–40/54)

At time t : **age** $A(t) = t - S_{N(t)}$, **remaining (residual) time** $Y(t) = S_{N(t)+1} - t$, **spread** $C(t) = X_{N(t)+1} = A(t) + Y(t)$. With $E[X^2] < \infty$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t A(s) ds = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t Y(s) ds = \frac{E[X^2]}{2E[X]}, \quad \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t C(s) ds = \frac{E[X^2]}{E[X]} \geq E[X]$$

(Proof 6). The interval covering a random time is **length-biased**: its limiting density is $xf(x)/E[X]$, and $E[X^2]/E[X] = E[X](1 + \text{cv}^2)$ exceeds $E[X]$ unless the gaps are deterministic — the **inspection paradox** (trains with headways 4, 12, 4, 12 min: a random passenger experiences a mean headway

of 10 min and waits 5 min, not 4). The limiting distribution of $A(t)$ and of $Y(t)$ is the **equilibrium distribution** $F_e(c) = \int_0^c (1 - F(x)) dx / E[X]$ (Proof 8). Bus stop (CP10 #4): passengers $PP(\lambda)$, bus gaps F : $L = \lambda E[X^2] / (2E[X])$ people waiting, $W = E[X^2] / (2E[X])$ mean wait, $L = \lambda W$. (*verified in notebook: CP10 §C, #4*)

1.7 Regenerative, alternating renewal, and semi-Markov processes (slides pp. 41–54/54)

A process is **regenerative** if there are epochs $T_1 < T_2 < \dots$ at which it restarts: the pieces of the path between successive T_n are iid copies, so the T_n form a renewal process. For a cycle T ,

$$\text{long-run fraction of time in state } j = \frac{E[\text{time in } j \text{ during a cycle}]}{E[T]} \quad (\text{Proof 7}),$$

which also equals $\lim_t P(X(t) = j)$ when the cycle length is non-lattice. An ergodic CTMC regenerates at every entry into a fixed state i , giving $p_i = (1/\lambda_i)/m_i$ with m_i the mean return time — the step Ch. 5 Proof 6(c) deferred to this chapter. A **semi-Markov process** (embedded DTMC P , general sojourn distributions with means μ_i) spends the long-run fraction $\pi_i \mu_i / \sum_j \pi_j \mu_j$ of its time in i (Proof 7). The **alternating renewal process** (on Z_n , off Y_n , iid pairs) is on the fraction $E[Z] / (E[Z] + E[Y])$ of the time (Proof 8): only the means enter (*insensitivity*) — availability of a machine, the busy fraction of the bank, the premium rates of CP10 #5 ($P_0 = e^{-\lambda s}$ at the discounted rate). (*verified in notebook: CP10 §E, #5*)

1.8 Model zoo (the class-problem cycles)

problem	renewal epoch / cycle X	$E[X]$	reward R per cycle and the answer
battery (CP9 #1)	a replacement; $X \sim \text{uniform}(50, 100)$	75 h	$R = 1$: rate $1/75$ per hour
bank (CP9 #2)	an entry; $X = \text{service} + \text{idle} \exp(\lambda)$	$\mu_G + 1/\lambda$	$R = 1$: entering rate $\lambda / (1 + \lambda \mu_G)$
door (CP10 #3)	a passage; $X = t + \exp(\lambda)$	$t + 1/\lambda$	$E[R] = \lambda t c + K e^{-\mu t} \lambda / (\lambda + \mu)$
age / residual	a renewal; $X \sim F$	$E[X]$	$R = X^2/2$: time average $E[X^2] / (2E[X])$
bus stop (CP10 #4)	a bus; $X \sim F$	$E[X]$	$R = \int_0^X N_p$, $E[R] = \lambda E[X^2]/2$: $L = \lambda E[X^2] / (2E[X])$
ergodic CTMC	an entry into i ; cycle mean m_i	m_i	$R = \text{time in } j$: p_j ; $p_i = (1/\lambda_i)/m_i$
on/off system	a switch-on; $X = Z + Y$	$E[Z] + E[Y]$	$R = Z$: on-fraction $E[Z] / (E[Z] + E[Y])$
age $\leq c$	a renewal; $X \sim F$	$E[X]$	$R = \min(X, c)$: $F_e(c) = \int_0^c (1 - F)/E[X]$
insurance (CP10 #5)	a claim; $X \sim \exp(\lambda)$	$1/\lambda$	$R = \min(s, X)$: $P_1 = 1 - e^{-\lambda s}$, average $r_0 e^{-\lambda s} + r_1 (1 - e^{-\lambda s})$

2 Core proofs

Notation: X_n iid $\sim F$ with $F(0) < 1$, $\mu = E[X]$, S_n , $N(t)$, $m(t)$ as above. Blanks are the steps *you* must supply in the gap-fill version.

Proof 1 — Counting from epochs: $N(t) \geq n \iff S_n \leq t$, the law of $N(t)$, and finiteness

Theorem. (a) $\{N(t) \geq n\} = \{S_n \leq t\}$, hence $P(N(t) = n) = F_n(t) - F_{n+1}(t)$ with F_n the cdf of S_n . (b) $N(t) < \infty$ almost surely for every t . (c) $m(t) = E[N(t)] < \infty$ for every t .

Proof. (a) “At least n renewals have occurred by time t ” means “the n -th renewal occurred at or before t ”: the two events are the same. Therefore $P(N(t) = n) = P(N(t) \geq n) - P(N(t) \geq n+1) = P(S_n \leq t) - P(S_{n+1} \leq t) = \underline{\hspace{2cm}}$. For exponential gaps $S_n \sim \text{Erlang}(n, \lambda)$, $F_n(t) = 1 - \sum_{i < n} e^{-\lambda t} (\lambda t)^i / i!$, and the difference telescopes to $e^{-\lambda t} (\lambda t)^n / n!$.

(b) By the strong law $S_n/n \rightarrow \mu > 0$ a.s., so $S_n \rightarrow \infty$ and only finitely many S_n are $\leq t$: $N(t) = \max\{n : S_n \leq t\} < \infty$. (If $\mu = \infty$, $S_n/n \rightarrow \infty$ still holds.)

(c) Since $F(0) < 1$ there is $\alpha > 0$ with $\beta := P(X > \alpha) > 0$. Truncate: $\bar{X}_n = \alpha \mathbf{1}\{X_n > \alpha\} \leq X_n$, with renewal process $\bar{N}(t) \geq N(t)$ (smaller gaps, more renewals). The truncated gaps take only the values 0 and α ; the number of gaps needed to accumulate one unit of α is geometric with mean $\frac{1}{\beta}$, and $\bar{N}(t)$ is at most the number of gaps needed to exceed t , i.e. to accumulate $\lfloor t/\alpha \rfloor + 1$ units:

$$m(t) \leq E[\bar{N}(t)] \leq \frac{\lfloor t/\alpha \rfloor + 1}{\beta} < \infty. \quad \square$$

(verified in notebook: CP9 §A (Erlang(2) gaps: pmf of $N(3)$ from Gamma cdfs vs simulation))

Checkpoint — answer before moving on.

- True or false, with a one-line reason (homework-style): $N(t) < n \iff S_n > t$; $N(t) \leq n \iff S_n \geq t$; $N(t) > n \iff S_n < t$.
- Why is “ $\geq$ ” in $\{N(t) \geq n\} = \{S_n \leq t\}$ essential — what goes wrong with $\{N(t) = n\} = \{S_n \leq t\}$?
- For Erlang(2, λ) gaps write $P(N(t) = n)$ as a combination of Poisson probabilities (every second event of a $\text{PP}(\lambda)$) and check it equals the Gamma-cdf difference.

AI study card — paste one of these into your AI assistant:

- Ask me to prove that $P(N(t) = n) = F_n(t) - F_{n+1}(t)$ for a renewal process, starting from the event identity between $N(t)$ and S_n . Then ask me which of three similar identities with strict/weak inequalities are true.
- Explain the truncation trick that shows $E[N(t)]$ is finite for any renewal process with $P(X = 0) < 1$, and quiz me on where $P(X = 0) < 1$ is used.

Proof 2 — The renewal function: tail sum, renewal equation, and two solutions (final-style)

Theorem. (a) $m(t) = \sum_{n \geq 1} F_n(t)$. (b) $m(t) = F(t) + \int_0^t m(t-x) dF(x)$. (c) Uniform(0, 1) gaps: $m(t) = e^t - 1$ for $0 \leq t \leq 1$. (d) Hyperexponential gaps, $F(t) = p(1 - e^{-\mu_1 t}) + (1-p)(1 - e^{-\mu_2 t})$:

$$m(t) = \frac{\mu_1 \mu_2}{c} t + \frac{p(1-p)(\mu_1 - \mu_2)^2}{c^2} (1 - e^{-ct}), \quad c = (1-p)\mu_1 + p\mu_2.$$

Proof. (a) For an integer-valued $N \geq 0$, $E[N] = \sum_{n \geq 1} P(N \geq n)$ (the event $\{N \geq n\}$ is counted once for each $n \leq N$). With $N = N(t)$ and Proof 1(a): $m(t) = \sum_n P(S_n \leq t) = \underline{\hspace{2cm}}$.

(b) Condition on X_1 : $m(t) = \int_0^\infty E[N(t) | X_1 = x] dF(x)$. If $x \leq t$ there is one renewal at x and afterwards a fresh renewal process on $(x, t]$, so $E[N(t) | X_1 = x] = \underline{\hspace{2cm}}$; if $x > t$ it is 0.

Hence $m(t) = \int_0^t (1 + m(t-x)) dF(x) = F(t) + \int_0^t m(t-x) dF(x)$.

(c) For $0 \leq t \leq 1$, $F(t) = t$ and $dF(x) = dx$ on $[0, t]$: $m(t) = t + \int_0^t m(t-x) dx = t + \int_0^t m(u) du$. Differentiating, $m'(t) = \underline{\hspace{2cm}}$ with $m(0) = 0$; $g = m + 1$ solves $g' = g$, $g(0) = 1$, so $m(t) = e^t - 1$.

(d) Laplace–Stieltjes transforms turn convolutions into products: $\tilde{F}_n(s) = \tilde{F}(s)^n$, so by (a) $\tilde{m}(s) = \sum_{n \geq 1} \tilde{F}(s)^n =$ _____ (equivalently: transform (b)). For the hyperexponential,

$$\tilde{F}(s) = \frac{p\mu_1}{s + \mu_1} + \frac{(1-p)\mu_2}{s + \mu_2} = \frac{s(p\mu_1 + (1-p)\mu_2) + \mu_1\mu_2}{(s + \mu_1)(s + \mu_2)},$$

and

$$1 - \tilde{F}(s) = \frac{(s + \mu_1)(s + \mu_2) - s(p\mu_1 + (1-p)\mu_2) - \mu_1\mu_2}{(s + \mu_1)(s + \mu_2)} = \frac{s(s + c)}{(s + \mu_1)(s + \mu_2)},$$

with $c =$ _____. Therefore

$$\tilde{m}(s) = \frac{s(p\mu_1 + (1-p)\mu_2) + \mu_1\mu_2}{s(s + c)} = \frac{A}{s} + \frac{B}{s + c},$$

$$A = \frac{\mu_1\mu_2}{c}, \quad B = p\mu_1 + (1-p)\mu_2 - \frac{\mu_1\mu_2}{c} = \text{_____}.$$

Since $\int_0^\infty e^{-st} d(At) = A/s$ and $\int_0^\infty e^{-st} d\left(\frac{B}{c}(1 - e^{-ct})\right) = B/(s + c)$, inversion gives $m(t) = At + \frac{B}{c}(1 - e^{-ct})$, which is the claim. $\square$

Remark. $A = \mu_1\mu_2/c = 1/E[X]$: the slope is the long-run rate (elementary renewal theorem), and $B/c = (\sigma^2 - \mu^2)/(2\mu^2) > 0$ here — a hyperexponential process starts “ahead” of the Poisson process with the same rate, the uniform(0, 1) one ($m(t) - 2t \rightarrow -\frac{1}{3}$) behind.

(verified in notebook: CP9 §B: grid solver vs $e^t - 1$ and vs the Laplace closed form)

Checkpoint — answer before moving on.

- **(final-style)** Compute $m(t)$ for $F(t) = p(1 - e^{-\mu_1 t}) + (1-p)(1 - e^{-\mu_2 t})$ from scratch, stating where the partial fractions come from. Then set $\mu_1 = \mu_2$ and check that you recover the Poisson process.
- Solve the renewal equation for Erlang(2, λ) gaps by transforms: $\tilde{F}(s) = (\lambda/(s + \lambda))^2$. (Answer: $m(t) = \lambda t/2 - (1 - e^{-2\lambda t})/4$.) Give a one-line probabilistic explanation via “every second event of a PP(λ)”.
- For uniform(0, 1) gaps and $1 \leq t \leq 2$ the equation becomes $m(t) = 1 + \int_{t-1}^t m(u) du$. Differentiate and solve it (use $m(u) = e^u - 1$ for $u \leq 1$).

AI study card — paste one of these into your AI assistant:

- Walk me through the Laplace-transform solution of the renewal equation for a hyperexponential inter-event distribution, one step at a time: the transform of F , $1 - F$ -tilde, the partial fractions, and the inversion. Do not show the result before I try each step.
- Give me a renewal equation $m(t) = F(t) + \int_0^t m(t-x) dF(x)$ with F uniform on $[0, 2]$. Ask me to turn it into a differential equation on $[0, 2]$ and to solve it, then check my answer numerically.

Proof 3 — The rate of a renewal process: $N(t)/t \rightarrow 1/\mu$ a.s. and $m(t)/t \rightarrow 1/\mu$

Theorem. (a) $N(t)/t \rightarrow 1/\mu$ almost surely. (b) (Elementary renewal theorem) $m(t)/t \rightarrow 1/\mu$.

Proof. (a) By definition $S_{N(t)} \leq t < S_{N(t)+1}$. Dividing by $N(t)$,

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)} = \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)}.$$

As $t \rightarrow \infty$, $N(t) \rightarrow \infty$ a.s. (every S_n is finite), so the strong law $S_n/n \rightarrow \mu$ applies along $n = N(t)$: the left side $\rightarrow \mu$ and the right side $\rightarrow \mu$. Hence $t/N(t) \rightarrow \mu$ a.s., i.e. $N(t)/t \rightarrow 1/\mu$.

(b) *Lower bound.* $N(t) + 1$ is a stopping time for the gaps (Proof 4), so Wald's equation gives $E[S_{N(t)+1}] = \mu(m(t) + 1)$. Since $S_{N(t)+1} > t$,

$$\mu(m(t) + 1) > t \implies \frac{m(t)}{t} > \frac{1}{\mu} - \frac{1}{t} \implies \liminf_{t \rightarrow \infty} \frac{m(t)}{t} \geq \frac{1}{\mu}.$$

Upper bound. Fix $M > 0$ and truncate the gaps: $\bar{X}_n = \min(X_n, M)$, with $\bar{\mu} = E[\bar{X}] \leq \mu$, $\bar{S}_n \leq S_n$, $\bar{N}(t) \geq N(t)$ and $\bar{m}(t) \geq m(t)$. Now $\bar{S}_{\bar{N}(t)+1} \leq t + M$ (the last gap is at most M), so Wald for the truncated process gives $\bar{\mu}(\bar{m}(t) + 1) \leq t + M$, i.e.

$$\frac{m(t)}{t} \leq \frac{\bar{m}(t)}{t} \leq \frac{t + M}{t \bar{\mu}} - \frac{1}{t} \xrightarrow{t \rightarrow \infty} \frac{1}{\bar{\mu}}.$$

Let $M \rightarrow \infty$: $\bar{\mu} \uparrow \mu$ (monotone convergence), so $\limsup_t m(t)/t \leq 1/\mu$. With the lower bound, $m(t)/t \rightarrow 1/\mu$. (For $\mu = \infty$ only the upper bound is needed, with $1/\bar{\mu} \downarrow 0$.) $\square$

Remark. (a) does not imply (b): almost-sure convergence of $N(t)/t$ says nothing about $E[N(t)]/t$ without a uniform integrability argument — which is what Wald plus truncation supplies. For the Poisson process $m(t)/t = \lambda$ exactly; in general the convergence is only asymptotic ($m(1) = e - 1 \approx 1.72$ versus $1/\mu = 2$ for uniform(0, 1) gaps).

(verified in notebook: CP9 §C–§D: $N(t)/t$ paths and Monte-Carlo $m(t)/t$ for three lifetime shapes; CP9 #1, #2)

Checkpoint — answer before moving on.

- **(final-style)** A battery lasts uniform(50, 100) hours and is replaced at once. State the long-run replacement rate and prove it from the sandwich inequality in four lines. Does the answer change for a lognormal lifetime with the same mean? What does change?
- **(final-style)** Single-server bank, Poisson(λ) arrivals accepted only when the server is free, service mean μ_G . Identify the cycle, compute the entering rate and the proportion entering, and say where the memoryless property is used. What if arrivals formed a renewal process with Erlang(2) gaps?
- In the upper bound, why is $\bar{S}_{\bar{N}(t)+1} \leq t + M$ true, and why does the same inequality fail for the untruncated process?

AI study card — paste one of these into your AI assistant:

- Make me prove $N(t)/t \rightarrow 1/\mu$ almost surely from $S_{\lfloor N(t) \rfloor} \leq t < S_{\lfloor N(t) \rfloor + 1}$ and the strong law, and then explain in one paragraph why this does not immediately give $m(t)/t \rightarrow 1/\mu$.
- Quiz me on the elementary renewal theorem: where Wald's equation enters, why $N(t)+1$ (not $N(t)$) is used, and why the upper bound needs truncation.

Proof 4 — Wald's equation and the corollary $E[S_{N(t)+1}] = \mu(m(t) + 1)$

Theorem. Let $X_1, X_2, \dots$ be iid with $E|X| < \infty$ and N a stopping time with $E[N] < \infty$. Then $E[\sum_{n=1}^N X_n] = E[N]E[X]$. Consequently, for a renewal process, $E[S_{N(t)+1}] = \mu(m(t) + 1)$.

Proof. Write the random sum with indicators, $\sum_{n=1}^N X_n = \sum_{n=1}^{\infty} X_n I_n$, $I_n = \mathbf{1}\{N \geq n\}$. The event $\{N \geq n\} = \{N \leq n-1\}^c$ and $\{N \leq n-1\} = \bigcup_{i=1}^{n-1} \{N = i\}$ is determined by $X_1, \dots, X_{n-1}$ (stopping time), so I_n is independent of $\underline{\quad}$. Then

$$E\left[\sum_{n=1}^N X_n\right] = \sum_{n=1}^{\infty} E[X_n I_n] = \sum_{n=1}^{\infty} E[X_n] E[I_n] = E[X] \sum_{n=1}^{\infty} P(N \geq n) = E[X] \underline{\quad},$$

where the interchange of E and $\sum$ is justified by monotone convergence for $X_n \geq 0$ and, for signed terms, by dominated convergence since $\sum_n E|X_n| E[I_n] = E|X| E[N] < \infty$ — this is where $E[N] < \infty$ is used; the last step is the tail-sum formula.

Corollary. $\{N(t) + 1 = n\} = \{S_{n-1} \leq t < S_n\}$ is determined by $X_1, \dots, X_n$, so $N(t) + 1$ is a stopping time, and $E[N(t) + 1] = m(t) + 1 < \infty$ (Proof 1(c)). Wald with $N = N(t) + 1$: $E[S_{N(t)+1}] = \mu(m(t) + 1)$. By contrast $\{N(t) = n\} = \{S_n \leq t < S_{n+1}\}$ depends on X_{n+1} : $N(t)$ is not a stopping time, and indeed $E[S_{N(t)}] \neq \mu m(t)$ in general. $\square$

(verified in notebook: CP9 §E: both identities checked by simulation; the random-walk and card examples)

Checkpoint — answer before moving on.

- (homework-style) A deck of 52 cards is turned over one at a time; $X_i = 1$ if the i -th card is an ace, and N is the position of the fourth ace. Is N a stopping time? Is Wald's equation valid for the X_i and N ? Compute $E[N] = 4 \cdot 53/5$ and $E[N]E[X_1]$ and compare with $\sum_{i \leq N} X_i = 4$.
- For the symmetric random walk and $N = \min\{n : S_n = 1\}$, which hypothesis of Wald fails? What does the equation " $1 = E[N] \cdot 0$ " tell you about $E[N]$?
- Give an example of a random index N that is *not* a stopping time for which Wald's equation nevertheless happens to hold.

AI study card — paste one of these into your AI assistant:

- Quiz me on the proof of Wald's equation: why $I_n = \mathbf{1}\{N \geq n\}$ is independent of X_n , where $E[N] < \infty$ is used, and why the tail-sum formula appears.
- Give me three random indices for an iid sequence (a hitting time, a last-exit time, a time chosen by a coin independent of the sequence) and ask me which are stopping times and for which Wald applies. Check my reasoning.

Proof 5 — The renewal reward theorem, and the door problem

Theorem. Let (X_n, R_n) be iid pairs with $E[X] < \infty$, $E[R] < \infty$, and $R(t) = \sum_{n=1}^{N(t)} R_n$. Then (a) $R(t)/t \rightarrow E[R]/E[X]$ a.s.; (b) $E[R(t)]/t \rightarrow E[R]/E[X]$.

Proof. (a) Write

$$\frac{R(t)}{t} = \frac{\sum_{n=1}^{N(t)} R_n}{N(t)} \cdot \frac{N(t)}{t}.$$

Since $N(t) \rightarrow \infty$ a.s., the strong law for the iid R_n gives $\sum_{n \leq N(t)} R_n / N(t) \rightarrow \underline{\quad}$, and Proof 3(a) gives $N(t)/t \rightarrow 1/E[X]$; the product converges to $E[R]/E[X]$. If rewards accrue continuously during

a cycle, the reward collected by t differs from $R(t)$ by at most the reward of the current cycle, and $R_{N(t)+1}/t = \frac{R_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{t} \rightarrow 0 \cdot \frac{1}{E[X]} = 0$ because $R_n/n \rightarrow 0$ a.s. for iid R_n with a finite mean.

(b) For $R_n \geq 0$: $N(t)+1$ is a stopping time for the pairs, so by Wald $E[\sum_{n=1}^{N(t)+1} R_n] = E[R](m(t)+1)$; dividing by t and using the elementary renewal theorem gives $E[R]/E[X]$ in the limit, and the extra term $E[R_{N(t)+1}]/t \rightarrow 0$ (it is bounded by $E[R \mathbf{1}\{\cdot\}]$ arguments we omit). Signed rewards: apply this to R_n^+ and R_n^- . $\square$

Application (CP10 #3). Door locked for t after each passage, Poisson(λ) arrivals, $\exp(\mu)$ service, cost c per customer lost at the door and K per customer who passes and finds the server busy. Cycle = from one passage to the next: $X = t + \exp(\lambda)$, $E[X] = t + 1/\lambda$. $E[C_1] = \lambda t c$ (Poisson(λt) arrivals during the lock). With $Y \sim \exp(\mu)$ the remaining service time at the cycle start — the same law whether or not the passing customer was served, by memorylessness — and $Z \sim \exp(\lambda)$ the wait after the lock, the next passer is lost iff $Y > t + Z$:

$$E[C_2] = K \int_0^\infty \lambda e^{-\lambda z} e^{-\mu(t+z)} dz = \frac{K \lambda e^{-\mu t}}{\lambda + \mu}, \quad \text{cost rate} = \frac{\lambda t c + K e^{-\mu t} \lambda / (\lambda + \mu)}{t + 1/\lambda}.$$

(verified in notebook: CP10 warm-up ($E[R]/E[X]$ vs $E[R/X]$), #3 (event-driven simulation vs the formula; the best lock length))

Checkpoint — answer before moving on.

- **(final-style)** Solve the door problem from scratch; then explain why the cycles are iid although the server may be busy when a cycle starts, and what would break with deterministic service times.
- Cycles $X \sim \text{uniform}(0.5, 1.5)$ with reward $R = X^2$: compute $E[R]/E[X]$ and $E[R/X]$. Which one is the long-run reward rate, and why?
- A reward R_n that depends on X_{n+1} (the *next* cycle) — does the theorem still apply? What about a reward depending on X_{n-1} ?

AI study card — paste one of these into your AI assistant:

- Ask me to prove the almost-sure half of the renewal reward theorem by splitting $R(t)/t$ into two factors, and to explain what happens to the reward of the unfinished last cycle.
- Give me a cost problem with a lockout time (customers lost at a locked door, a server that may be busy) and make me identify the cycle, $E[X]$, $E[R]$ and the cost rate. Then change the service distribution and ask me what fails.

Proof 6 — Age and remaining time, the inspection paradox, and the bus stop

Theorem. Let $E[X^2] < \infty$. (a) $\frac{1}{t} \int_0^t A(s) ds \rightarrow \frac{E[X^2]}{2E[X]}$ and $\frac{1}{t} \int_0^t Y(s) ds \rightarrow \frac{E[X^2]}{2E[X]}$ a.s.; hence the time average of the spread $C = A + Y$ is $E[X^2]/E[X]$. (b) The long-run fraction of time during which $C(t) \leq x$ is $\int_0^x y dF(y)/E[X]$ (length-biased law of the covering interval). (c) Bus stop: passengers PP(λ), bus gaps F : $L = \lambda E[X^2]/(2E[X])$, $W = E[X^2]/(2E[X])$, $L = \lambda W$.

Proof. (a) The age rises at slope 1 from 0 at each renewal, so during the n -th cycle the area under it is $R_n = \int_0^{X_n} s ds = \frac{X_n^2}{2}$ — iid rewards, each depending on its own cycle, with $E[R] = E[X^2]/2 < \infty$.

Split at the last renewal before t :

$$\frac{1}{t} \int_0^t A(s) ds = \frac{\sum_{n=1}^{N(t)} R_n}{t} + \frac{\int_{S_{N(t)}}^t A(s) ds}{t},$$

the first term $\rightarrow E[X^2]/(2E[X])$ by Proof 5, the second is at most $X_{N(t)+1}^2/(2t) \rightarrow 0$ (the last-cycle argument of Proof 5). The remaining time falls at slope 1 to 0, with the same area $X_n^2/2$ per cycle, so its time average is the same; adding, the time average of C is $E[X^2]/E[X]$.

(b) Reward $R_n = X_n \mathbf{1}\{X_n \leq x\}$ = time during cycle n at which the covering interval has length $\leq x$. By Proof 5 the fraction of time with $C(t) \leq x$ is $E[X \mathbf{1}\{X \leq x\}]/E[X] = \frac{\int_0^x f(x) dx}{E[X]}$: a law with density $xf(x)/E[X]$ when F has a density, and mean $E[X^2]/E[X] = E[X] + \sigma^2/E[X] \geq E[X]$ — long intervals cover more of the time axis, so a random time falls into a long interval more often (inspection paradox). Equality iff $\sigma = 0$.

(c) Buses are renewal epochs and every bus takes everyone waiting. In a cycle the number waiting s time units after a bus is a Poisson count $N_p(s)$ with mean λs (Poisson passengers, independent of the buses), so with $R = \int_0^X N_p(s) ds$,

$$E[R] = \int_0^\infty E\left[\int_0^x N_p(s) ds\right] dF(x) = \int_0^\infty \int_0^x \lambda s ds dF(x) = \text{_____} \quad (\text{Fubini}),$$

so $L = E[R]/E[X] = \lambda E[X^2]/(2E[X])$. Given one passenger arrival in $(0, t]$, its time is uniform on $(0, t)$ and independent of the buses, so a passenger arrives at a random time of the bus process and waits its remaining time: W = time average of $Y = E[X^2]/(2E[X])$ by (a). Comparing, $L = \lambda W$ — Little's law, here from two renewal reward arguments. $\square$

Numbers. $\lambda = 2/\text{min}$, bus gaps uniform(5, 15): $E[X^2] = 108.3$, $W = 5.42$ min, $L = 10.8$; deterministic gaps of 10: $W = 5$, $L = 10$; exponential gaps of mean 10: $W = 10$, $L = 20$. Trains with headways 4, 12, 4, 12: mean headway seen 10 min, mean wait 5 min.

(verified in notebook: CP10 §C (sawtooth, length-biased histogram, cv experiment), #4 (direct simulation, Little check))

Checkpoint — answer before moving on.

- **(final-style)** Derive L and W for the bus stop and verify $L = \lambda W$. Which step uses “Poisson arrivals see time averages”?
- For exponential gaps (a) gives mean age = mean remaining = $1/\lambda$. Reconcile with memorylessness, and explain why the spread then averages $2/\lambda$ although every gap has mean $1/\lambda$.
- The friendship paradox: in a network with degrees d_i , a friend of a random person has on average $\sum_i d_i^2 / \sum_i d_i$ friends. Map this onto (b): what is “length” and what is the “random time”?

AI study card — paste one of these into your AI assistant:

- Make me derive the time average of the age of a renewal process with the renewal reward theorem (reward = area under the sawtooth in one cycle). Then ask what changes for the remaining time and for the spread.
- Explain the inspection paradox with a bus schedule whose headways alternate 4 and 12 minutes; make me compute the mean headway a random passenger experiences and the mean wait, and then generalise to $E[X^2]/E[X]$.

Proof 7 — Regenerative processes: time fractions, the ergodic CTMC, and semi-Markov processes

Theorem. (a) For a regenerative process with cycle length T , $E[T] < \infty$, the long-run fraction of time in state j is $E[\text{time in } j \text{ during a cycle}]/E[T]$ (and equals $\lim_t P(X(t) = j)$ when T is non-lattice). (b) For an ergodic CTMC, $p_i = (1/\lambda_i)/m_i$ where m_i is the mean return time to i . (c) For a semi-Markov process with embedded chain P (stationary distribution π) and mean sojourns μ_i , the fraction of time in i is $\pi_i \mu_i / \sum_j \pi_j \mu_j$.

Proof. (a) Let R_n = time spent in j during the n -th cycle. The cycles are iid copies, so the pairs (cycle length, R_n) are iid and Proof 5 gives $\frac{1}{t} \int_0^t \mathbf{1}\{X(s) = j\} ds \rightarrow \frac{E[R_n]}{E[T]}$ a.s. (The identification with $\lim_t P(X(t) = j)$ is the key renewal theorem, not proved here; for a lattice T the probability may oscillate while the time fraction exists.)

(b) By the Markov property and the exponential sojourns, the past is forgotten at every entry into i : the successive i -to- i excursions are iid cycles with mean length m_i . Within a cycle the time in i is one sojourn, mean μ_i , so $p_i = (1/\lambda_i)/m_i$ by (a). Equivalently $m_i = 1/(\lambda_i p_i)$: entries into i occur at long-run rate $\lambda_i p_i$ (Ch. 5 Proof 6(c), HW3 A3(v)).

(c) Regenerate at the entries into a fixed state i . Between two visits to i the embedded chain visits j on average π_j/π_i times (the standard DTMC identity: stationary mass is proportional to the expected number of visits per i -cycle), each visit lasting μ_j on average and the sojourns being independent of the route; so the expected time in j per cycle is $\pi_j \mu_j / \pi_i$ and the expected cycle length is $\sum_k \pi_k \mu_k / \pi_i$. Divide: fraction in j = $\frac{\pi_j \mu_j}{\sum_k \pi_k \mu_k}$. $\square$

Remarks. (b) does not need the sojourns to be exponential: any iid cycle structure works, which is why availability-type answers survive non-exponential clocks (HW3 Part C). A CTMC is the semi-Markov process with $\mu_i = 1/\lambda_i$, and (c) then reads $p_i \propto \pi_i / \lambda_i$ — Ch. 5 Proof 6(d) again.

(verified in notebook: CP10 §E (machine up/down: cycle formula vs *machine_repair*; six on/off shapes))

Checkpoint — answer before moving on.

- Two machines share one repair person; up times $\exp(\mu)$, repair times deterministic. Is the number of working machines a regenerative process? Name a regeneration epoch. Is *one machine's* up/down sequence an alternating renewal process? (HW3 Part C(b) explained.)
- A semi-Markov process on $\{0, 1, 2\}$ cycles $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ with sojourn means 1, 2, 3. Compute the time fractions by (c) and by a direct cycle argument.
- For the machine with $\exp(\mu)$ up and $\exp(\lambda)$ repair, compute p_{up} by (b) with $i = \text{up}$, and compare with $p_Q = 0$.

AI study card — paste one of these into your AI assistant:

- Explain regenerative processes to me with a queue that empties, then quiz me on why the fraction of time in a state is a ratio of expectations and when it also equals a limiting probability.
- Give me a 3-state semi-Markov process (embedded matrix and sojourn means) and make me compute the long-run time fractions, then make me recover the CTMC case by setting the means to $1/\lambda_i$.

Proof 8 — The alternating renewal process, the age process (final-style), and the insurance rates

Theorem. (a) For an on/off system with iid pairs (Z_n, Y_n) of on and off times, $E[Z], E[Y] < \infty$, the long-run fraction of time on is $E[Z]/(E[Z] + E[Y])$. (b) For a renewal process with gaps $X \sim F$ and any $c > 0$,

$$\text{long-run fraction of time with } A(t) \leq c = \frac{\int_0^c (1 - F(x)) dx}{E[X]} =: F_e(c),$$

and the same for $Y(t)$; F_e has mean $E[X^2]/(2E[X])$. (c) Insurance (CP10 #5): claims $\text{PP}(\lambda)$, rate r_1 until s claim-free years then r_0 until the next claim: $P_1 = 1 - e^{-\lambda s}$, $P_0 = e^{-\lambda s}$, average premium $r_0 e^{-\lambda s} + r_1(1 - e^{-\lambda s})$.

Proof. (a) The switch-on epochs form a renewal process with cycles $X_n = Z_n + Y_n$; take the reward $R_n = Z_n$ (time on during the cycle), which depends on its own cycle only. Proof 5: fraction on $= E[Z]/E[Z + Y] = \frac{E[Z]}{E[X]}$. Note that Z_n and Y_n may be dependent — only the *pairs* must be iid.

(b) Call the system on at time t if $A(t) \leq c$. Within a renewal interval of length X the age is $\leq c$ for exactly $\min(X, c)$ time units and $> c$ for the rest, so this is an alternating renewal process with cycle X and on-time $Z = \min(X, c)$. For the nonnegative variable $\min(X, c)$ the tail formula gives

$$E[\min(X, c)] = \int_0^\infty P(\min(X, c) > x) dx = \int_0^c P(X > x) dx = \frac{\int_0^c (1 - F(x)) dx}{E[X]},$$

and (a) gives the fraction $\int_0^c (1 - F(x)) dx / E[X]$. Running each cycle backwards turns the age into the remaining time, so $Y(t)$ has the same long-run law. The mean of F_e : by the tail formula and Fubini,

$$\int_0^\infty (1 - F_e(c)) dc = \frac{1}{E[X]} \int_0^\infty \int_c^\infty (1 - F(x)) dx dc = \frac{1}{E[X]} \int_0^\infty x(1 - F(x)) dx = \frac{E[X^2]}{2E[X]},$$

using $\int_0^\infty x(1 - F(x)) dx = E[X^2]/2$ (integration by parts).

(c) A claim resets the rate to r_1 , and since claims form a Poisson process the time to the next claim is $\exp(\lambda)$ whatever happened before: the claim epochs are regeneration points and a cycle is $X \sim \exp(\lambda)$, $E[X] = 1/\lambda$. The customer pays r_1 for $\min(s, X)$ and r_0 for $(X - s)^+$. By the tail formula, $E[\min(s, X)] = \int_0^s e^{-\lambda x} dx = \frac{1 - e^{-\lambda s}}{\lambda}$, so $P_1 = E[\min(s, X)]/E[X] = 1 - e^{-\lambda s}$, $P_0 = e^{-\lambda s}$, and the long-run average paid per year is $r_0 P_0 + r_1 P_1$. $\square$

Remarks. Only the means of the on/off times enter (a): the availability $m_u/(m_u + m_r)$ of a machine is *insensitive* to the shapes of the up and repair time distributions — HW3 Part C(a),(c). What is sensitive is a machine that must *wait* for a shared repair person: its down time then depends on the other machine, so its own up/down pairs are not iid (Proof 7 checkpoint). Part (b) with $F = \exp(\lambda)$ gives $F_e = F$ (the age of a Poisson process is $\exp(\lambda)$); with uniform(0, 1), $F_e(c) = 2c - c^2$ — the age is $\leq \frac{1}{4}$ for 43.75% of the time although only 25% of the intervals are that short. Part (c) is also (b) with $c = s$ applied to the claim process.

(verified in notebook: CP10 §E (six shapes, age $\leq c$ check), #5 (claims simulation; s and λ sliders))

Checkpoint — answer before moving on.

- **(final-style)** Show that the proportion of time the age of a renewal process is less than c equals $\int_0^c (1 - F(x)) dx / E[X]$, using the alternating renewal process and the identity $E[Y] = \int_0^\infty (1 - G(y)) dy$ for $Y \geq 0$ with cdf G .

- **(final-style)** Solve the insurance problem; then redo (a) with the claim-free requirement counted from the *start of the policy* only once (the discount, once earned, is kept until the next claim — as stated) versus a rule where every claim restarts the s -year clock. Are the cycles different?
- A machine has uniform(1,3) up times and exp(1) repair times. Compute the availability. Why can $pQ = 0$ not be used, and which result replaces it?

AI study card — paste one of these into your AI assistant:

- Ask me to derive the long-run fraction of time that the age of a renewal process is at most c , by identifying an on/off structure, and to check it for exponential and for uniform gaps.
- Give me an on/off system where the on and off times of a cycle are dependent (e.g. off = part of the same random variable). Ask me whether the formula $E[Z]/(E[Z]+E[Y])$ still applies and why.

Where to next. CP10 wrap-up → these proofs (gap-fill first) → Checkpoint quiz at the start of the next lecture → HW 4 Part A (closed-book style, no AI on the first attempt) → Part C: how the gap variance sets the convergence speed, and which of HW3's insensitivity findings the cycle argument explains.